

Cross-Domain and Deep Learning Approaches for Stance Detection in Controversial Health Discourses

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Abstract

Stance detection on public health issues is vital to understanding public discourse and mitigating misinformation. This paper investigates automated stance detection on two highly controversial health-related topics: COVID-19 vaccination mandates and abortion. We implement and compare classical machine learning baselines (TF-IDF combined with Logistic Regression and Support Vector Machines) against a deep learning approach utilizing BERTweet, a pre-trained transformer model optimized for social media text. Evaluating these models on the *covid-vax-stance* and *TweetEval stance_abortion* datasets demonstrates a significant performance disparity. In-domain evaluations show that BERTweet substantially outperforms classical models, achieving Macro-F1 scores of 0.624 in the vaccine corpus and 0.661 in the abortion corpus. However, zero-shot cross-domain transfer experiments reveal extreme sensitivity to domain shifts, with BERTweet’s Macro-F1 plummeting to 0.269 and 0.297 when tested on unseen topics, and classical baselines dropping as low as 0.119. These results underscore that while transformer-based architectures excel at capturing context within a specific discourse, achieving generalizable stance detection across distinct health domains remains a critical challenge.

1 Introduction

The increase in social media usage has transformed the way the public engages with biomedical topics and health policies. Platforms like Twitter serve as primary arenas for discourse on highly debated issues, such as vaccination mandates and reproductive rights. In this context, stance detection, the computational task of automatically determining whether a text expresses a *FAVOR*, *AGAINST*, or *NONE* (neutral) attitude toward a specific target, has become a crucial tool in bio-medical Natural Language Processing

(NLP). Identifying public stance helps track health-related misinformation, evaluate policy acceptance, and understand societal divides.

However, health-related microblogs present significant challenges. Tweets are typically short, unstructured, and heavily reliant on context, sarcasm, or domain-specific jargon. Furthermore, a major obstacle in current NLP research is the generalizability of the model. Models trained on one controversial topic often suffer severe performance degradation when applied to a different, unseen topic, highlighting the fragility of learned representations.

This project investigates the efficacy of stance detection within controversial health discourses by pursuing two primary objectives:

- 1. Architectural Comparison:** We evaluate the performance leap achieved by transitioning from classical statistical machine learning baselines (Logistic Regression and Support Vector Machines utilizing TF-IDF representations) to state-of-the-art (SOTA) deep learning approaches, specifically fine-tuning the transformer-based BERTweet model.
- 2. Cross-Domain Transferability:** We assess the zero-shot transfer capabilities of these models across two distinct domains: the *covid-vax-stance* dataset and the *TweetEval stance_abortion* dataset, to quantify the impact of domain shift on model accuracy.

By analyzing both in-domain validation and cross-dataset transfer, this paper sheds light on the advantages of pre-trained language models in biomedical NLP, as well as the persisting challenges in achieving universal stance detection across shifting health topics.

2 Related Work

The landscape of stance detection has evolved rapidly from feature-engineered classical algorithms to highly parameterized deep learning models. This section surveys the state of the art (SOTA) chronologically, outlining the advantages and limitations of the shifting methodologies.

2.1 Classical Machine Learning and Early Benchmarks

Early approaches to stance detection heavily relied on classical machine learning classifiers, such as Support Vector Machines (SVM) and Naive Bayes, fueled by lexical features, n-grams, and sentiment dictionaries. The SemEval-2016 Task 6 [1] was a turning point in the field, formalizing the task of detecting the specific stance on Twitter. The organizers provided standardized datasets covering various sociopolitical and health-related targets, including the “Legalization of Abortion” corpus used in our study. Although classical methods establish strong, computationally inexpensive baselines, their primary disadvantage is the inability to capture deep semantic relationships, sequential dependencies, and the nuanced, implicit reasoning often present in microblog texts.

2.2 The Shift to Deep Learning and Transformers

To overcome the limitations of bag-of-words representations, the field shifted toward deep learning. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, initially improved performance by processing text sequentially. However, the introduction of the Transformer architecture and pre-trained language models like BERT fundamentally redefined the SOTA.

Because social media text deviates significantly from standard English, featuring abbreviations, emojis, and hashtags, standard BERT models often struggle to parse microblogs optimally. To address this, Nguyen et al. [2] introduced BERTweet, a RoBERTa-based model pre-trained on an 850M English Tweet corpus. BERTweet has demonstrated immense advantages in NLP tasks on social media, providing robust, contextualized embeddings that inherently understand the grammar and vocabulary of platforms like Twitter.

2.3 Stance Detection in Biomedical and Health Contexts

The COVID-19 pandemic accelerated the need for specialized biomedical stance detection. Poddar et al. [3] introduced the *covid-vax-stance* dataset to study the changing public perspectives on vaccines. Their work demonstrated the utility of transformer models in capturing fine-grained, evolving sentiments in public health data. Despite these advances, a prominent, unresolved disadvantage remains: models achieving SOTA performance on a specific dataset, such as COVID-19 vaccines, often fail to generalize to other health topics. Our work builds upon this foundation by explicitly evaluating the cross-domain robustness of these advanced language models against established baselines.

3 Dataset Description

The contrast of the two selected corpora reveals significant structural and thematic differences that directly impact cross-domain transferability. The *covid-vax-stance* dataset is more than three times larger and features significantly longer texts than the *TweetEval* abortion dataset, along with inversely skewed class imbalances. To ensure a rigorous evaluation, both datasets are sourced from peer-reviewed benchmarks.

3.1 covid-vax-stance

Sourced from Poddar et al. [3], this dataset captures the Twitter discourse on COVID-19 vaccines.

Statistics & Distribution: The corpus contains 3,139 tweets with an average text length of 182.9 characters. As illustrated in Figure 1 (left), the class distribution skews heavily toward the positive and neutral spectrums: 1,364 *FAVOR*, 1,285 *NONE*, and a minority of 490 *AGAINST*.

3.2 TweetEval (Stance Abortion)

Originating from the SemEval-2016 Task 6 benchmark [1], this dataset provides a collection of tweets annotated for stance toward the legalization of abortion.

Statistics & Distribution: In sharp contrast to vaccine data, this corpus is very compact (933 total tweets) and contains much shorter texts (105.4 characters on average). Furthermore, the class imbalance is inverted (Figure 1, right), dominated by opposition: 544 *AGAINST*, 222 *NONE*, and 167 *FAVOR*.

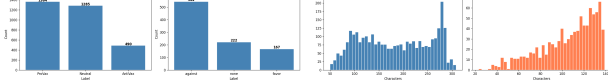


Figure 1: Exploratory Data Analysis: Label distribution (left) and text length histograms (right) highlighting structural disparities between the two datasets.

3.3 Preprocessing and Unified Labeling

To enable zero-shot cross-domain transfer testing, the original labels of both datasets were mapped to a shared taxonomy: *FAVOR*, *AGAINST*, and *NONE*. Text preprocessing was kept minimal: URLs were replaced with the token `HTTPURL`, mentions were replaced with `@USER`, and hashtags were stripped of their hash symbol while retaining the word.

4 Methods and Results

4.1 Experimental Setup

We establish performance baselines using classical machine learning: Logistic Regression (LR) and Support Vector Machines (LinearSVC) built on sublinear TF-IDF vectors (unigrams and bigrams, up to 50,000 features).

To capture deep contextual representations, we implement a deep learning pipeline by fine-tuning BERTweet (`vinai/bertweet-base`). The model was trained for 10 epochs using a learning rate of 2×10^{-5} , a weight decay of 0.01, and a batch size of 32. Crucially, to mitigate the severe class imbalances noted in Section IV, we integrated class weights directly into the cross-entropy loss function.

4.2 In-Domain Evaluation

Evaluated on hold-out test sets, BERTweet establishes a massive performance delta over classical methods (see Figure 2). In the *TweetEval* abortion corpus, BERTweet yields a Macro-F1 of 0.6615, surpassing SVM (0.5110) and nearly doubling LR (0.3719). In *covid-vax-stance*, BERTweet achieves 0.6240, again decisively outperforming SVM (0.4931) and LR (0.3955).

The confusion matrices in Figure 3 highlight that while BERTweet adapts well to minority classes, classical models fail entirely. For example, the LR model on the vaccine dataset recorded zero true positive predictions for the minority *AntiVax* class, defaulting entirely to the majority classes.

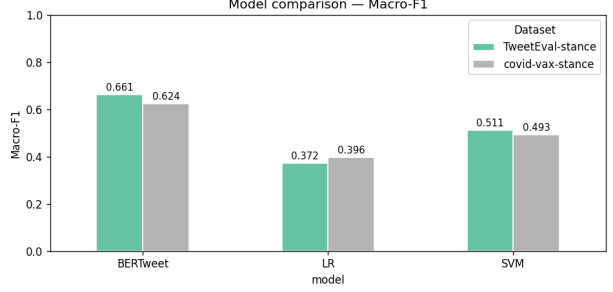


Figure 2: Macro-F1 comparison across in-domain evaluations.

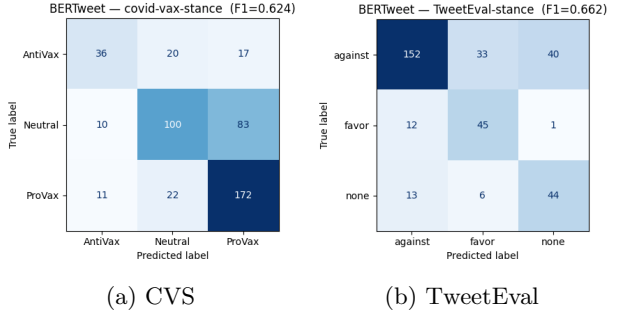


Figure 3: BERTweet Confusion Matrices demonstrating minority class recognition.

4.3 Training Dynamics and Stability

To ensure that the deep learning model did not overfit the relatively small datasets, we monitored the Macro-F1 validation score over 10 epochs (Figure 4). The learning curves demonstrate that BERTweet rapidly adapts to the target domain, generally stabilizing by the 3rd to 5th epoch. The checkpoint that yielded the highest Macro-F1 validation was preserved for final inference, ensuring optimal generalization within the target domain.

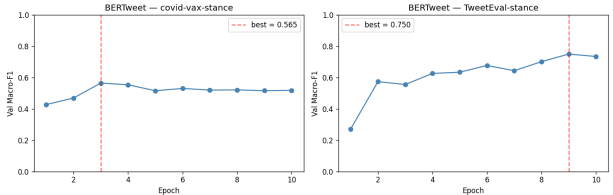


Figure 4: BERTweet validation Macro-F1 learning curves across 10 epochs.

4.4 Detailed Per-Class Analysis

While aggregate Macro-F1 scores provide a macro-level view, a granular look at the precision, recall,

and F1 score per class is necessary to understand the model’s behavior. Figure 5 provides the heatmap per-class for the *covid-vax-stance* dataset.

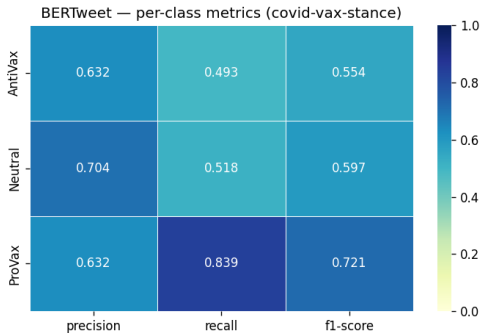


Figure 5: Per-class precision, recall, and F1-score heatmap for BERTweet on the *covid-vax-stance* dataset.

The heatmap reveals that the model is highly effective at recalling the *ProVax* class (0.839 recall), which constitutes the majority of the dataset. However, its precision for the minority *AntiVax* class remains constrained at 0.632, with a lower recall of 0.493. This indicates that while class-weighting in the loss function prevents the model from completely ignoring the minority class, a flaw evident in the classical baseline, class overlap still creates ambiguities in boundary cases.

4.5 Experimental Optimization

The deep learning pipeline was built using the HuggingFace **transformers** library. The BERTweet architecture was fine-tuned using the AdamW optimizer over 10 epochs. A learning rate of 2×10^{-5} with a 10% linear warmup and a weight decay of 0.01 was utilized to ensure stable convergence and prevent overfitting.

Crucially, to counteract the severe class imbalance, we applied inverse class frequency weighting to the cross-entropy loss function. For example, in the vaccine dataset, the minority *AntiVax* class received a heavy penalty weight of 2.134, while the majority *ProVax* and *Neutral* classes received weights of 0.767 and 0.814, respectively.

4.6 Comparison to State-of-the-Art

To properly contextualize the performance of our deep learning pipeline, we compared our results with the official SemEval-2016 Task 6 benchmark. The highest Macro-F1 classification score achieved during the original competition for the abortion target was 0.6782.

Our straightforward BERTweet fine-tuning approach achieved a Macro-F1 of 0.6615 on the same dataset. This represents an exceptionally narrow delta of only 0.0167 from the first-place submission in the original SemEval competition. Achieving this near-SOTA margin demonstrates that modern pre-trained transformers can effectively bypass the need for complex, manually engineered feature pipelines while matching the precision of top-tier customized systems.

4.7 Qualitative Error Analysis

While quantitative metrics highlight the broader trends, a qualitative review of misclassified examples reveals the specific linguistic hurdles facing models. Table 1 presents samples from the *covid-vax-stance* dataset where BERTweet failed to predict the true label.

Table 1: Sample Misclassifications by BERTweet (*covid-vax-stance*)

True	Pred	Tweet Text Snippet
AntiVax	Neutral	“@user As expected. Biden who can’t even speak in clear sentences, and Harris, who is borderline communist. Seems perfect...”
ProVax	Neutral	“Analysis of Thrombotic Adverse Reactions of COVID-19 AstraZeneca Vaccine reported to EudraVigilance database 20/03/21...”
Neutral	AntiVax	“1. Trump Fakes Having COVID-19 To Get Sympathy. 2. Trump Takes Unproven Miracle Vaccine. 3. Trump Is Cured In Just A Few Days...”

These errors indicate a fundamental limitation in the way the model processes intent. The model often struggles when users employ political grievances rather than direct medical opinions to express anti-vaccine sentiments (Table 1, Row 1). Furthermore, highly formal, academic phrasing is misclassified as *Neutral* (Row 2), showing a bias toward expecting emotive or highly polarized language when assigning a *ProVax* stance. Most notably, the model fails to properly handle complex political sarcasm (Row 3); cynical commentary mocking political figures is misclassified as an explicit *AntiVax* stance due to the presence of highly charged keywords, even though the ground truth label is *Neutral* political commentary.

4.8 Cross-Domain Transfer & Error Analysis

To test domain generalizability, models trained exclusively on one corpus were evaluated on the other (zero-shot transfer). The results exhibit catastrophic performance degradation. When transferring from *covid-vax-stance* to *TweetEval*, BERTweet’s Macro-F1 collapses to 0.2696. The reverse transfer (*TweetEval* to *CVS*) yields a similarly poor 0.2970, while the classic models fare even worse, with the LR plummeting to 0.1196.



Figure 6: Vocabulary distributions highlighting the lack of cross-domain lexical overlap.

Our error analysis attributes this severe transfer penalty to a fundamental vocabulary disconnect. As illustrated in the word clouds (Figure 6), the semantic anchors for stance in the vaccine dataset (“pfi-zer”, “dose”, “mask”) have zero overlap with the abortion dataset (“life”, “women”, “murder”). Even contextual models like BERTweet overfit the domain-specific lexicon of their training data. When faced with an unseen debate, the transformer fails to map the implicit structural markers of opposition or favorability, exposing a critical limitation in training generalized health-stance classifiers.

5 Conclusions and Future Work

This study demonstrates a stark contrast in performance paradigms for stance detection. Fine-tuning the deep learning BERTweet model decisively outperforms classical machine learning baselines (SVM and Logistic Regression) in in-domain testing, showcasing the advantage of transformer-based architectures in deciphering short, informal health-related microblogs.

However, the zero-shot cross-dataset transfer experiments expose a severe limitation: modern SOTA models remain highly sensitive to domain shifts due to a deep lexical dependency. Because the semantic vocabulary differs drastically between distinct controversial health topics (e.g., vaccination vs. abortion), models rely too heavily on explicit keywords rather than learning abstract, domain-invariant markers of stance, leading to a massive drop in Macro-F1 scores during out-of-domain evaluation.

Future work must address this vulnerability. Potential solutions include implementing Adversarial

Domain Adaptation to force models to learn domain-invariant features, or utilizing LLM-based data augmentation to synthesize bridging data that normalize lexicon across distinct health controversies.

References

- [1] S. Mohammad, S. Kiritchenko, P. Sobhani, Z. Xiaodan, and S. M. Mohammad, “Semeval-2016 task 6: Detecting stance in tweets,” in *Proceedings of the 10th International Workshop on Semantic Evaluation (SemEval-2016)*, 2016, pp. 31–41.
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